ISTQB CERTIFIED TESTER FOUNDATION LEVEL V4.0

# Question Bank & Mock Exams

152 exam-style multiple-choice questions: 72 chapter drills with immediate answers and explanations, plus two full 40-question mock exams mirroring the official question distribution - each with a consolidated answer key and rationale for every item.

**72 Chapter Questions · 2 Full Mock Exams**

K1 REMEMBER · K2 UNDERSTAND · K3 APPLY

# Table of Contents

How to Use This Question Bank 2

Part 1 - Chapter Drills (72 questions with answers) 2

- Drill 1 - Fundamentals of Testing . . . 2
- Drill 2 - Testing Throughout the Software Development Lifecycle . . . 5
- Drill 3 - Static Testing . . . 8
- Drill 4 - Test Analysis and Design . . . 11
- Drill 5 - Managing the Test Activities . . . 14
- Drill 6 - Test Tools . . . 17

Part 1 - Mock Exam 1 (40 questions, 60 minutes) 20

Part 2 - Mock Exam 2 (40 questions, 60 minutes) 27

Part 4 - Mock Exam Answer Keys 34

- Mock Exam 1 - Answer Grid . . . 34
- Mock Exam 2 - Answer Grid . . . 34
- Mock Exam 1 - Rationales . . . 34
- Mock Exam 2 - Rationales . . . 37

# How to Use This Question Bank

This document contains 152 original multiple-choice questions written strictly against the CTFL v4.0 syllabus. Part 1 provides twelve chapter drills per chapter with the answer and explanation directly beneath each question - use them on the study day dedicated to that chapter, with your notes closed. Parts 2 and 3 are complete 40-question mock exams that mirror the official question distribution across the six chapters; sit them under real exam conditions of 60 minutes with no interruptions. Part 4 holds the consolidated answer keys with a rationale for every mock question.

Two rules make the drills work. First, treat every guessed answer as wrong even when it is right - guesses reveal lucky spots that will not survive the real exam. Second, for every miss, write in your error notebook why the correct option is correct AND why the tempting option exists; ISTQB distractors are engineered from true-sounding syllabus sentences placed in the wrong context. A mock score of 26 or above passes, 30 is comfortable, and 32 or more is your target before exam day.

# Part 1 - Chapter Drills (72 questions with answers)

Answer each drill question before looking at the answer box. The K-level tag shows the cognitive level of the matching syllabus learning objective: K1 questions test recall of terms and concepts, K2 questions test understanding and comparison, and K3 questions test application of techniques. When you finish a chapter drill, review every miss immediately and record it in your error notebook before moving on.

## Drill 1 - Fundamentals of Testing

**1. [KK1]** Which statement correctly describes a typical test objective?

**A.** Proving that the software contains no remaining defects
**B.** Evaluating work products such as requirements, designs, and code
**C.** Replacing the need for quality assurance activities
**D.** Exhaustively executing every possible input combination

> **Answer: B** - Evaluating work products is a listed test objective. Testing shows presence, not absence, of defects; it does not replace QA; exhaustive testing is impossible.

**2. [KK2]** During dynamic testing, a failure was observed. Which sequence describes the typical debugging process that follows?

**A.** Fix the cause, reproduce the failure, then diagnose the root cause
**B.** Reproduce the failure, diagnose the root cause, then fix the cause
**C.** Diagnose the root cause, add new tests, then re-run the full regression suite
**D.** Report the failure, remove the defect, then perform static analysis

> **Answer: B** - The debugging process is: reproduction of the failure, diagnosis (finding the root cause), fixing the cause. Confirmation testing follows.

**3. [KK2]** A defect is found during a review (static testing) of the design document. Why does debugging this defect not require reproduction and diagnosis of a failure?

**A.** Because static testing cannot find real defects
**B.** Because static testing finds defects directly, without triggering a failure
**C.** Because design defects are always cheaper to fix than code defects
**D.** Because reviews are performed by the author who already knows the cause

> **Answer: B** - Static testing finds defects directly. Dynamic testing triggers failures that must then be reproduced and diagnosed; static defects have no failure to reproduce.

**4. [KK2]** Which statement best distinguishes quality assurance (QA) from quality control (QC)?

**A.** QA is product-oriented and corrective; QC is process-oriented and preventive
**B.** QA is process-oriented and preventive; QC is product-oriented and corrective
**C.** Both QA and QC focus exclusively on executing tests against code
**D.** QA applies only to testing processes, QC applies only to development processes

> **Answer: B** - QA is process-oriented and preventive; QC is product-oriented and corrective, and testing is a major form of QC. QA applies to both development and testing processes.

**5. [KK2]** A developer is tired after a long shift and writes a division that can receive zero as the denominator. The tester executes the division with zero and the system crashes. Match the terms to this scenario.

**A.** Failure: tiredness; error: the crash; defect: the zero input
**B.** Error: tiredness led to the defect (missing guard); failure: the crash during execution
**C.** Defect: tiredness; failure: the missing guard; error: the crash
**D.** Root cause: the crash; defect: tiredness; failure: the missing guard

> **Answer: B** - Human error (tiredness) produced a defect in the code; executing the defective code caused the failure (crash). The root cause is the situation that led to the error.

**6. [KK1]** Radiation at a customer site flips bits in firmware, causing the system to behave incorrectly, although no human made any coding mistake. What does this illustrate?

**A.** Failures require a human error somewhere in the lifecycle
**B.** Failures can be caused by environmental conditions, not only by human errors
**C.** The radiation caused an error which is therefore the root cause of a defect in the specification
**D.** This is a defect without a failure and therefore out of scope of testing

> **Answer: B** - Errors and defects are not the only causes of failures; environmental conditions such as radiation or electromagnetic fields can cause failures too.

**7. [KK2]** A project manager asks the team to test every possible combination of inputs for a billing system. Based on the testing principles, what is the best response?

**A.** Agree, because exhaustive testing is the only way to satisfy principle one

**B.** Explain that exhaustive testing is impossible and propose test techniques, test case prioritization, and risk-based testing to focus the effort

**C.** Agree, but only for the components with the fewest parameters

**D.** Refuse, because testing everything is always a waste of the available budget

> **Answer: B** - Principle 2: exhaustive testing is impossible; instead use techniques, prioritization, and risk-based testing to focus test efforts.

**8. [KK2]** Historical data shows that 80 percent of the defects of a similar product were found in three modules. How should this influence testing?

**A.** Spread the effort evenly, because defects cannot be predicted

**B.** Use the predicted defect clusters as an important input for risk-based testing and focus effort on those modules

**C.** Test only those three modules and skip the rest of the system

**D.** Repeat the same tests across all modules until they wear out

> **Answer: B** - Principle 4 (defects cluster together, a Pareto illustration): predicted and observed clusters are an important input for risk-based testing - focus, not skip.

**9. [KK2]** After running the same manual test cases for the fourth release, the team finds almost no new defects, although defect rates elsewhere are unchanged. What does this illustrate and what is the recommended response?

**A.** Principle 2; increase the number of executions of the same tests

**B.** Principle 5, tests wear out; modify existing tests and write new ones, while keeping beneficial repetition such as automated regression testing

**C.** Principle 6; change the development lifecycle because testing is context dependent

**D.** Principle 1; testing no longer shows the presence of defects, so stop testing

> **Answer: B** - Tests wear out with repetition. The response is to modify tests and test data and add new tests; automated regression repetition remains beneficial.

**10. [KK2]** A team verifies every specified requirement, fixes all found defects, and still the delivered system does not fulfill the users' actual needs or the customer's business goals. Which principle does this scenario demonstrate?

**A.** Testing is context dependent

**B.** Absence-of-defects fallacy: verification alone is not enough, validation is also required

**C.** Defects cluster together

**D.** Early testing saves time and money

> **Answer: B** - Principle 7: expecting verification to ensure system success is a fallacy; a defect-free system can still fail users' needs - validation must also be performed.

**11. [KK2]** Which test process activity answers the question 'what to test?' in terms of measurable coverage criteria?

**A.** Test design
**B.** Test analysis
**C.** Test implementation
**D.** Test control

> **Answer: B** - Test analysis analyzes the test basis to identify testable features and define test conditions - what to test. Test design answers how to test.

**12. [KK2]** What is the primary value of maintaining traceability between the test basis and the testware?

**A.** It guarantees that all defects will be found before release
**B.** It supports coverage evaluation, impact analysis of changes, and audit requirements
**C.** It removes the need for a test completion report
**D.** It allows testers to skip test design for low-risk features

> **Answer: B** - Traceability supports evaluating coverage, analyzing which tests are affected by a change, and audit trail needs - it guarantees nothing by itself.

## Drill 2 - Testing Throughout the Software Development Lifecycle

**1. [KK2]** How does the choice of the software development lifecycle MOST directly affect testing?

**A.** It has no effect as long as the test techniques are chosen correctly
**B.** It affects the scope and timing of test activities, documentation detail, techniques, automation extent, and the tester's role
**C.** It only changes the number of test levels, never the test types
**D.** It determines that dynamic testing must always be performed first

> **Answer: B** - The SDLC impacts scope/timing of activities, documentation detail, technique choice, automation extent, and tester responsibilities.

**2. [KK1]** Which of the following is a good testing practice regardless of the SDLC model?

**A.** Waiting for the code to be complete before writing any test case
**B.** For every development activity there is a corresponding test activity, and test analysis for a level begins during the corresponding development phase
**C.** Using one single test level for the whole project to avoid redundancy
**D.** Reviewing work products only after formal delivery to testing

> **Answer: B** - The good practices include a test activity for every development activity, distinct objectives per level, early test analysis and design, and early reviewer involvement.

**3. [KK1]** Which of the following are examples of test-first approaches to development?

**A.** Waterfall, V-model, spiral model
**B.** TDD, ATDD, and BDD
**C.** Kanban, Lean IT, and FDD
**D.** Retrospectives, daily scrum, and backlog refinement

> **Answer: B** - TDD, ATDD, and BDD define tests before the code is written, implementing early testing and shift-left. The other lists are lifecycle models or Agile practices without the test-first essence.

**4. [KK2]** In Behavior-Driven Development (BDD), how is the desired behavior of the application expressed?

**A.** As UML state diagrams translated into code by the IDE
**B.** As test cases written in a simple natural-language form, usually Given/When/Then, automatically translated into executable tests
**C.** As unit tests written by developers after the code is refactored
**D.** As acceptance criteria in a tabulated input-output mapping only

> **Answer: B** - BDD expresses behavior with test cases in simple natural language (Given/When/Then) that stakeholders understand, then translates them into executable tests.

**5. [KK2]** Which of the following is a testing benefit of DevOps?

**A.** It removes the need for manual testing from the user's perspective
**B.** Fast feedback on code quality and minimized regression risk through large automated regression suites
**C.** It guarantees stable requirements before development starts
**D.** It eliminates the need to define a delivery pipeline

> **Answer: B** - DevOps benefits include fast feedback, CI-driven shift-left, stable test environments, non-functional attention, less repetitive manual testing, and minimized regression risk. Manual user-perspective testing is still needed.

**6. [KK2]** Which practice is an example of the shift-left approach?

**A.** Performing all non-functional testing after the complete system is delivered
**B.** Reviewing specifications from a testing perspective and writing test cases before the code is written
**C.** Merging all testing into a single system test phase at the end
**D.** Delaying static analysis until after dynamic testing finishes

> **Answer: B** - Shift-left practices include specification reviews, test cases before code, CI/CD usage, early static analysis, and starting non-functional testing at component level where possible.

**7. [KK2]** Which test level focuses on testing the interfaces and interactions between components, heavily dependent on bottom-up, top-down, or big-bang strategies?

**A.** Component testing
**B.** Component integration testing
**C.** System testing
**D.** System integration testing

> **Answer: B** - Component integration testing (unit integration testing) targets interfaces and interactions between components and depends on the chosen integration strategy.

**8. [KK2]** Which of the following lists ONLY forms of acceptance testing?

**A.** User acceptance, operational, contractual and regulatory, alpha, beta
**B.** Smoke, sanity, exploratory, usability
**C.** Component, component integration, system, system integration
**D.** Functional, non-functional, black-box, white-box

> **Answer: B** - The main forms of acceptance testing are UAT, operational acceptance, contractual and regulatory acceptance, alpha testing, and beta testing.

**9. [KK2]** According to ISO/IEC 25010 as cited in the syllabus, which of the following is a non-functional software quality characteristic?

**A.** Functional correctness
**B.** Portability
**C.** Test completeness
**D.** Defect density

> **Answer: B** - The ISO/IEC 25010 characteristics listed are performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

**10. [KK2]** A defect was fixed. The team re-runs the previously failed test cases and confirms the fix works. What additional testing should follow, and what does it check?

**A.** Confirmation testing; it checks that the fix works in production
**B.** Regression testing; it checks that the change caused no adverse consequences elsewhere
**C.** Acceptance testing; it checks the user's business needs
**D.** System integration testing; it checks interfaces with external services

> **Answer: B** - Re-running failed cases is confirmation testing. Regression testing then confirms the change broke nothing in the same component, other components, or connected systems.

**11. [KK2]** Why is regression testing a strong candidate for automation?

**A.** Because regression tests are written only once and never change
**B.** Because regression suites run many times and grow with each iteration or release
**C.** Because automation removes the need for impact analysis
**D.** Because manual regression testing is prohibited in DevOps

> **Answer: B** - Regression suites are executed repeatedly and their case count grows each iteration, so automation should start early in the project; with CI, automated regression tests are included in the pipeline.

**12. [KK2]** A company migrates its application from an on-premise platform to a cloud platform, including converting legacy data. Under which maintenance trigger does this testing fall?

**A.** Modification - a planned enhancement
**B.** Upgrade or migration of the operational environment, including data conversion testing
**C.** Retirement - end of life of the application
**D.** Confirmation testing of a hot fix

> **Answer: B** - Environment upgrades/migrations are one of the three trigger groups (modifications, upgrades/migrations, retirement) and include testing the new environment and data conversion.

## Drill 3 - Static Testing

**1. [KK1]** Which of the following work products can be examined by static testing?

**A.** Only executable source code
**B.** Almost any work product that can be read and understood, including requirements, test plans, and contracts
**C.** Only documents with a formal syntax
**D.** Only third-party executable code submitted for analysis

> **Answer: B** - Any readable, understandable work product can be reviewed. Static analysis additionally needs a structure to check against, such as formal-syntax code or models.

**2. [KK2]** Why is static testing considered valuable even though reviews can be costly to implement?

**A.** It replaces the need for dynamic testing of the code
**B.** It detects defects in the earliest phases, finds defects dynamic testing cannot, and usually lowers overall project costs
**C.** It guarantees that all requirements are correct without stakeholder involvement
**D.** It is the only technique able to measure performance efficiency

> **Answer: B** - Static testing fulfills early testing, finds unreachable code and non-executable-product defects, builds shared understanding, and reduces total cost despite review costs.

**3. [KK2]** Which characteristic can be measured by static testing but NOT by dynamic testing?

**A.** Performance efficiency
**B.** Maintainability of the code
**C.** Response time
**D.** Reliability under load

> **Answer: B** - Static testing measures quality characteristics not dependent on executing code (e.g. maintainability); dynamic testing measures those that depend on execution (e.g. performance efficiency).

**4. [KK1]** What is a benefit of early and frequent stakeholder feedback during the SDLC?

**A.** It eliminates all requirement changes before development begins

**B.** It prevents misunderstandings about requirements and keeps the team focused on the most valuable features

**C.** It allows the team to skip reviews of user stories

**D.** It transfers quality responsibility from the team to the stakeholders

> **Answer: B** - Frequent feedback prevents requirement misunderstandings, ensures changes are understood and implemented earlier, and focuses effort on the features of most value and risk impact.

**5. [KK2]** Which sequence of review process activities is correct?

**A.** Planning, individual review, review initiation, fixing and reporting, communication and analysis

**B.** Planning, review initiation, individual review, communication and analysis, fixing and reporting

**C.** Review initiation, planning, communication and analysis, individual review, fixing and reporting

**D.** Individual review, planning, review initiation, fixing and reporting, communication and analysis

> **Answer: B** - The ISO/IEC 20246 generic order is: planning, review initiation, individual review, communication and analysis, fixing and reporting.

**6. [KK2]** During the communication and analysis activity, reviewers have logged 12 anomalies. What is the correct handling of these anomalies?

**A.** All anomalies are defects and must be fixed before the meeting ends

**B.** Anomalies are not necessarily defects; each gets a decision on status, ownership, and required actions

**C.** Anomalies are discarded unless the author agrees they are real

**D.** Anomalies are only recorded when found during individual review, never in the meeting

> **Answer: B** - Anomalies identified during a review are not necessarily defects; the meeting decides status, ownership, and actions, and also assesses the work product's quality level.

**7. [KK1]** In a review, who decides what is to be reviewed and provides time, staff, and other resources?

**A.** The review leader

**B.** The manager

**C.** The moderator

**D.** The scribe

> **Answer: B** - The manager decides what is reviewed and provides resources. The review leader organizes who, when, and where; the moderator runs the meeting; the scribe records.

**8. [KK1]** Which review role is responsible for collating anomalies from reviewers and recording decisions and new anomalies found during the review meeting?

**A.** The moderator
**B.** The author
**C.** The scribe
**D.** The reviewer

> **Answer: B** - The scribe (recorder) collates anomalies and records review information such as decisions and new anomalies found during the meeting.

**9. [KK2]** Which statement about a walkthrough is correct?

**A.** It is led by a moderator and requires technically qualified reviewers
**B.** It is led by the author and serves objectives such as education, consensus, and confidence building; individual review beforehand is optional
**C.** It follows the complete generic review process and collects metrics
**D.** It prohibits the author from acting as review leader or scribe

> **Answer: B** - A walkthrough is led by the author, serves many objectives (quality evaluation, confidence, education, consensus, ideas, author improvement, anomaly detection), and pre-session individual review is not required.

**10. [KK2]** What distinguishes an inspection from the other review types?

**A.** It is the least formal type and needs no documented output
**B.** It is the most formal type, follows the complete generic process, aims to find the maximum number of anomalies, collects metrics, and the author cannot be review leader or scribe
**C.** It is led by the author to educate reviewers and build consensus
**D.** It is reserved exclusively for source code reviews

> **Answer: B** - Inspections are the most formal, follow the complete generic process, target the maximum number of anomalies, and use metrics to improve the SDLC; the author cannot act as review leader or scribe.

**11. [KK1]** Which of the following is a success factor for reviews?

**A.** Using reviews to evaluate the participants of the review
**B.** Conducting reviews on small chunks so concentration is maintained, with clear objectives and measurable exit criteria
**C.** Reviewing an entire large work product in a single six-hour session
**D.** Skipping preparation so that reviewers judge with fresh eyes

> **Answer: B** - Success factors include small chunks, clear objectives with measurable exit criteria, feedback to stakeholders and authors, adequate preparation time, management support, culture, and training. Evaluating participants must never be an objective.

**12. [KK2]** Which types of defects are typically found more cheaply by static testing than by dynamic testing?

**A.** Memory leaks that appear only after 48 hours of load
**B.** Requirements ambiguities, deviations from standards, and unreachable code
**C.** Concurrency failures under production-like traffic
**D.** Incorrect response time under peak usage

> **Answer: B** - Static testing excels at requirements inconsistencies/ambiguities, design and coding defects, standard deviations, interface specification errors, security vulnerabilities such as buffer overflows, and coverage gaps.

## Drill 4 - Test Analysis and Design

**1. [KK3]** A web form accepts an order quantity from 1 to 500 inclusive. Which set of equivalence partitions is correct?

**A.** Valid: 1-500; invalid: none, because the form validates input
**B.** Valid: 1-500; invalid: values below 1 and values above 500
**C.** Valid: 0-500; invalid: 501 and above
**D.** Valid: all integers; invalid: all non-integers

> **Answer: B** - The valid partition is the specified range 1-500; the invalid partitions are $x < 1$ and $x > 500$. Invalid partitions must be tested as well as valid ones.

**2. [KK3]** Using equivalence partitioning for a single ordered input with one valid and two invalid partitions, how many test cases achieve 100% partition coverage?

**A.** One test with a value from the valid partition
**B.** Three tests, one from each partition
**C.** One test per boundary value
**D.** It is impossible to reach 100% with invalid partitions

> **Answer: B** - Coverage items in EP are the partitions; 100% coverage requires exercising every identified partition - valid and invalid - at least once, so one test per partition.

**3. [KK3]** For the input range 1-500, which values are tested by 2-value boundary value analysis?

**A.** 1, 2, 499, 500
**B.** 0, 1, 2, 499, 500, 501
**C.** 0, 1, 500, 501
**D.** 1, 250, 500

> **Answer: C** - 2-value BVA tests each boundary and its closest neighbor from the adjacent partition: for boundaries 1 and 500 that is 0, 1, 500, 501. The extra neighbors 2 and 499 belong to 3-value BVA.

**4. [KK3]** The requirement says if (x <= 10) but the code implements if (x == 10). Why can 3-value BVA detect this defect while 2-value BVA may not?

**A.** 3-value BVA tests only invalid partitions, where the defect is visible

**B.** 3-value BVA adds the second neighbor (x = 9), which is likely to expose the wrongly narrowed condition

**C.** 3-value BVA replaces boundary testing with random values

**D.** 2-value BVA does not test boundary values at all

> **Answer: B** - 2-value BVA gives 10 and 11, both passing the wrong code; 3-value BVA adds 9, which satisfies x <= 10 but not x == 10, exposing the defect.

**5. [KK3]** A business rule depends on three Boolean conditions. How many columns does the FULL decision table contain before simplification?

**A.** 3

**B.** 6

**C.** 8

**D.** 9

> **Answer: C** - Rules grow exponentially: 2 to the power of the number of conditions. With three Boolean conditions the full table has 2^3 = 8 columns.

**6. [KK3]** In a limited-entry decision table, a condition cell contains a dash (-). What does this notation mean?

**A.** The condition is infeasible for this rule

**B.** The value of the condition is irrelevant for the outcome of this rule

**C.** The condition must be tested in a separate test case

**D.** The condition is always true in this rule

> **Answer: B** - T = satisfied, F = not satisfied, dash = irrelevant for the outcome, N/A = infeasible for the given rule.

**7. [KK3]** A state machine has 6 states. A test suite visits all 6 states and exercises all valid transitions. Which statement is correct?

**A.** All states coverage is achieved; valid transitions coverage is achieved and guarantees full all states coverage

**B.** All transitions coverage is achieved, because all states were visited

**C.** Valid transitions coverage is achieved but all states coverage is still incomplete

**D.** None of the coverage criteria can be claimed without testing invalid transitions

> **Answer: B** - Full valid-transitions coverage guarantees full all-states coverage. All-transitions coverage additionally requires attempting invalid transitions.

**8. [KK3]** When testing invalid transitions in state transition testing, why should a single test case attempt only ONE invalid transition?

**A.** To keep the test execution schedule within the time-box
**B.** To avoid fault masking, where one defect prevents the detection of another
**C.** Because state diagrams do not show invalid transitions
**D.** Because invalid transitions cannot occur in the same state twice

> **Answer: B** - Testing one invalid transition per test case avoids fault masking - a situation in which one defect prevents the detection of another.

**9. [KK2]** A test suite achieves 100% branch coverage of a component. What else is guaranteed?

**A.** 100% statement coverage, because branch coverage subsumes statement coverage
**B.** 100% valid-transitions coverage of the related state machine
**C.** Detection of all data-dependent defects such as division by zero
**D.** Verification that every requirement was implemented

> **Answer: B** - Branch coverage subsumes statement coverage: 100% branch coverage always delivers 100% statement coverage, but not vice versa. Requirement omissions are a white-box weakness.

**10. [KK2]** What is a fundamental weakness shared by all white-box techniques?

**A.** They cannot be applied until after design or implementation
**B.** They may not detect defects of omission when the software fails to implement a requirement
**C.** They provide no objective measure of code coverage
**D.** They are unusable when the specification is vague or outdated

> **Answer: B** - White-box testing takes the whole implementation into account - strong with vague specs - but if a requirement is not implemented at all, structural testing may miss the omission. They also only exist after design/implementation.

**11. [KK2]** A tester has two hours, no specification, and an unstable build. Which experience-based technique fits BEST, and why?

**A.** Checklist-based testing, because checklists replace the missing specification
**B.** Exploratory testing, because tests are designed, executed, and evaluated simultaneously while learning the object, within a time-boxed session
**C.** Error guessing, because fault attacks require a full requirement list
**D.** State transition testing, because the model can be built during execution

> **Answer: B** - Exploratory testing is the technique of choice with few or inadequate specifications and time pressure, often structured as session-based testing with a charter and debriefing.

**12. [KK2]** In ATDD, after the specification workshop and the positive test cases, what is the recommended order of remaining coverage?

**A.** Non-functional tests first, then negative tests
**B.** Negative tests, then non-functional quality characteristics
**C.** Random order, as long as all acceptance criteria are covered
**D.** Automation code, then negative tests, then the specification

> **Answer: B** - The ATDD order is: first positive test cases, then negative testing, finally non-functional characteristics such as performance efficiency and usability.

## Drill 5 - Managing the Test Activities

**1. [KK2]** Which item belongs to the typical content of a test plan?

**A.** The full list of defects found in the previous release
**B.** The test approach, including test levels, types, techniques, and entry and exit criteria
**C.** The signed acceptance certificate of the system
**D.** The production monitoring dashboard configuration

> **Answer: B** - Test plan content includes context, assumptions and constraints, stakeholders, communication, risk register, test approach (levels, types, techniques, deliverables, entry/exit criteria, independence, metrics, data and environment), budget, and schedule.

**2. [KK1]** In iteration planning, which contribution is expected from testers?

**A.** Defining the organizational test policy for the whole company
**B.** Participating in detailed risk analysis of user stories and breaking them into tasks, including testing tasks
**C.** Signing the release contract with the customer
**D.** Replacing the product owner in prioritizing the product backlog

> **Answer: B** - Iteration planning: detailed risk analysis, testability checks, creating acceptance tests, breaking stories into tasks, estimating testing tasks, and identifying functional and non-functional aspects. Release planning covers testable stories, risk analysis, effort estimation, approach, and release test planning.

**3. [KK2]** In Agile development, what are the Agile names for exit criteria of a releasable item and entry criteria for a user story?

**A.** Definition of Ready and Definition of Done, respectively
**B.** Definition of Done and Definition of Ready, respectively
**C.** Test charter and test session sheet
**D.** Sprint backlog and product backlog

> **Answer: B** - Exit criteria are called Definition of Done (metrics for a releasable item); the entry criteria a user story must fulfill to start development/testing are the Definition of Ready.

**4. [KK3]** Using three-point estimation with a = 6, m = 9, b = 18 person-hours, what is the final estimate E and its standard deviation SD?

**A.** E = 10, SD = 2
**B.** E = 9, SD = 3
**C.** E = 11, SD = 6
**D.** E = 10, SD = 12

> **Answer: A** - E = (a + 4m + b) / 6 = (6 + 36 + 18) / 6 = 10; SD = (b - a) / 6 = 12 / 6 = 2. Report as 10 plus or minus 2 person-hours.

**5. [KK3]** In a previous project the ratio of development effort to test effort was 3:2. The current project is estimated at 600 person-days of development. Using estimation based on ratios, what test effort should be planned?

**A.** 300 person-days
**B.** 400 person-days
**C.** 600 person-days
**D.** 900 person-days

> **Answer: B** - With a 3:2 ratio, test effort = 600 x (2/3) = 400 person-days. Ratios from the organization's own historical projects are the best source.

**6. [KK3]** A high-priority test case can only run after a low-priority test case has prepared its environment. How should the test execution schedule handle this?

**A.** Skip the low-priority case because priority rules override everything
**B.** Execute the lower-priority test case first, because dependencies override priority order
**C.** Merge both cases into a single automated script before deciding
**D.** Escalate to the test manager to re-rank the risks before execution

> **Answer: B** - Ideal ordering follows priority, but dependencies force the lower-priority prerequisite to run first; resource availability windows also constrain the order.

**7. [KK1]** According to the test pyramid model, which statement about the layers is correct?

**A.** Higher layers contain faster, more isolated tests
**B.** Bottom-layer tests are small, isolated, fast, and usually needed in large numbers, while top-layer tests are complex, slow end-to-end tests needed in small numbers
**C.** All layers should contain the same number of tests to balance coverage
**D.** The pyramid forbids automation at the top layer

> **Answer: B** - The higher the layer, the lower the granularity, isolation, and execution time; the bottom needs many small fast tests, the top a few complex end-to-end tests.

**8. [KK2]** Which tests belong to Quadrant Q3 (business facing, critique the product)?

**A.** Component and component integration tests
**B.** Exploratory testing, usability testing, and user acceptance testing
**C.** Smoke tests and performance tests
**D.** API testing and simulations

> **Answer: B** - Q3 holds user-oriented, often manual tests: exploratory, usability, UAT. Q1 = automated component/integration in CI; Q2 = functional tests, examples, story tests, prototypes, API testing; Q4 = smoke and non-functional except usability.

**9. [KK2]** Which of the following is a PRODUCT risk rather than a project risk?

**A.** A third-party supplier fails to deliver a required component
**B.** Incorrect tax calculations that could cause financial penalties for users
**C.** Two key testers resign during the project
**D.** The test environment delivery is delayed by two weeks

> **Answer: B** - Product risks concern quality characteristics of the product itself, e.g. wrong calculations with consequences like criminal penalties. The other options are organizational, people, and supplier project risks.

**10. [KK1]** Risk likelihood is 0.2 and risk impact is rated 40 on a monetary scale. Using the quantitative approach, what is the risk level?

**A.** 40.2
**B.** 8
**C.** 42
**D.** 200

> **Answer: B** - Quantitatively, risk level = likelihood x impact = 0.2 x 40 = 8. Qualitatively, the level would instead be read from a risk matrix.

**11. [KK2]** Which content is typical of a test PROGRESS report rather than a test completion report?

**A.** Lessons learned relevant to the testing
**B.** Testing planned for the next period, current impediments, and new or changed risks
**C.** Deviations from the test plan in schedule and effort
**D.** Evaluation of product quality against the original exit criteria

> **Answer: B** - Progress reports (periodic) contain the test period, progress and deviations, impediments, metrics, new/changed risks, and planned testing. Completion reports summarize the stage: summary, evaluation vs. plan, deviations, unmitigated risks, unfixed defects, lessons learned.

**12. [KK3]** A tester logs a defect. Which information set makes the defect report MOST useful for reproduction and resolution?

**A.** Identifier, title, and the author's opinion about who caused it

**B.** Identifier, title, date and author, test object and environment, context, reproduction steps with logs, expected and actual results, severity, priority, status, references

**C.** Identifier, title, and a screenshot only; the rest can be asked later

**D.** Identifier, the failing requirement, and the fix proposal

> **Answer: B** - The full defect report structure includes identifier, title, date/organization/author with role, test object and environment, context, failure description with steps and evidence, expected vs. actual results, severity, priority, status, and references.

## Drill 6 - Test Tools

**1. [KK2]** Which tool category MOST directly supports reviews and static analysis?

**A.** Management tools

**B.** Static testing tools

**C.** DevOps tools

**D.** Collaboration tools

> **Answer: B** - Static testing tools support the tester in performing reviews and static analysis; management tools handle SDLC/requirements/tests/defects/configuration management.

**2. [KK2]** A team wants automated generation of test cases, test data, and test procedures. Which tool category fits?

**A.** Test design and implementation tools

**B.** Test execution and coverage tools

**C.** Non-functional testing tools

**D.** Tools supporting scalability and standardization

> **Answer: A** - Test design and implementation tools facilitate the generation of test cases, test data, and test procedures; execution and coverage tools run tests and measure coverage.

**3. [KK2]** Which activity is supported by test execution and coverage tools?

**A.** Automated test execution and coverage measurement

**B.** Communication between distributed team members

**C.** Containerization of the test environment

**D.** Workflow tracking of the delivery pipeline

> **Answer: A** - Execution and coverage tools automate running tests and measure coverage; DevOps tools handle pipeline and builds; collaboration tools handle communication.

**4. [KK1]** Which statement about test tool acquisition is correct?

**A.** Buying the market-leading tool guarantees lasting benefits

**B.** Simply acquiring a tool does not guarantee success; benefits require effort for introduction, maintenance, and training

**C.** Tools pay for themselves within one sprint in most organizations

**D.** Tool risk disappears once the tool is integrated into CI

> **Answer: B** - The syllabus is explicit: acquiring a tool guarantees nothing; real and lasting benefits require effort for introduction, maintenance, and training, and risks need analysis and mitigation.

**5. [KK1]** Which of the following is a potential BENEFIT of test automation?

**A.** Complete elimination of the testing team

**B.** Time saved by reducing repetitive manual work such as regression execution and result comparison

**C.** Removal of the need for test management reporting

**D.** Guaranteed detection of every defect in the product

> **Answer: B** - Benefits: less repetitive manual work, greater consistency and repeatability, more objective assessment, easier access to testing information, faster execution and feedback, and more time for deeper test design.

**6. [KK1]** Which of the following is a potential RISK of test automation?

**A.** More objective assessment of coverage

**B.** Unrealistic expectations about tool functionality and ease of use, and dependency on a vendor that may fail or retire the tool

**C.** Reduced test execution time

**D.** Prevention of simple human errors through repeatability

> **Answer: B** - Risks include unrealistic expectations, underestimated costs, using tools where manual testing fits better, over-reliance, vendor dependency, open-source abandonment, platform incompatibility, and unsuitable tools for regulatory needs.

**7. [KK1]** A company automates a regression suite, then stops designing new tests because 'the tool now covers everything'. Which automation risk does this illustrate?

**A.** Using an open-source tool that was abandoned

**B.** Relying on a tool too much, ignoring the need for human critical thinking

**C.** Choosing a tool incompatible with the development platform

**D.** Inaccurate estimation of the time required to introduce the tool

> **Answer: B** - Over-reliance on a tool - ignoring human critical thinking - is a listed risk; automation complements but never replaces test design thinking.

**8. [KK2]** Non-functional testing tools are valuable because they:

**A.** Replace the need for a representative test environment

**B.** Allow the tester to perform non-functional testing that is difficult or impossible to perform manually

**C.** Convert non-functional requirements into functional ones

**D.** Automatically fix performance defects found during testing

> **Answer: B** - The syllabus defines non-functional testing tools as enabling testing that is difficult or impossible manually, e.g. load generation and measurement.

**9. [KK1]** Which example shows an unsuitable tool choice from a compliance perspective?

**A.** A spreadsheet used to track test execution in a small project

**B.** An automation tool that does not comply with the regulatory requirements and safety standards applicable to the product

**C.** A container tool used to standardize deployment of test environments

**D.** A management tool used to track requirements and defects

> **Answer: B** - Choosing an unsuitable tool that did not comply with regulatory requirements and/or safety standards is a listed risk. A spreadsheet is explicitly acceptable as a test tool in context.

**10. [KK2]** Which statement about automation and manual testing is correct?

**A.** Mature automation removes the need for manual testing entirely

**B.** Even with a high level of automated testing, manual testing - especially from the user's perspective - is still needed

**C.** Manual testing is only needed when tools are unavailable

**D.** Automation and manual testing must never be mixed within one test level

> **Answer: B** - The syllabus states that despite high automation (e.g. in DevOps), manual testing from the user's perspective remains necessary.

**11. [KK2]** Which pairing of tool category to activity is correct?

**A.** DevOps tools - generating unit test data from models

**B.** Management tools - facilitating management of requirements, tests, defects, and configuration

**C.** Collaboration tools - measuring code coverage percentages

**D.** Static testing tools - deploying builds to production

> **Answer: B** - Management tools increase test process efficiency by facilitating SDLC, requirements, test, defect, and configuration management. The other pairings mix categories.

**12. [KK1]** How does automation through a delivery pipeline affect repetitive manual work and information access?

**A.** It increases manual repetitions while reducing data visibility

**B.** It reduces repetitive manual work and provides easier access to statistics, graphs, and aggregated data for reporting

**C.** It has no measurable effect on either

**D.** It removes the need for test data preparation of any kind

> **Answer: B** - Automation reduces repetitive manual testing and improves access to testing information - statistics, graphs, aggregated progress, defect rates, and durations - supporting management and reporting.

# Part 1 - Mock Exam 1 (40 questions, 60 minutes)

Simulate the real exam completely: 40 questions, 60 minutes on the clock, closed book, no pauses, no phone. Answer every question - there is no negative marking - and flag anything that takes longer than ninety seconds for a second pass. Questions appear in syllabus order; the distribution matches the official CTFL v4.0 exam: 8 questions from Chapter 1, 7 from Chapter 2, 4 from Chapter 3, 13 from Chapter 4, 6 from Chapter 5, and 2 from Chapter 6. Record your answers on paper, then score with the key in Part 4. Pass mark: 26 of 40 (65 percent).

**1.** Testing is BEST described as:

**A.** Only executing the software and checking results

**B.** A set of activities to discover defects and evaluate the quality of software artifacts, including static activities and planning

**C.** A process whose only purpose is to verify that requirements are implemented

**D.** An activity performed exclusively by an independent test team at the end of the project

**2.** Checking whether the system meets the users' needs in its operational environment is called:

**A.** Verification

**B.** Validation

**C.** Debugging

**D.** Static analysis

**3.** Which statement about the relationship between testing and debugging is correct?

**A.** Testing and debugging are the same activity performed by different people

**B.** Dynamic testing triggers failures; debugging finds and eliminates the causes (defects)

**C.** Debugging must always be performed before any test execution starts

**D.** Static testing results always require failure reproduction before fixing

**4.** A defect in a requirements specification is not detected and is used to design the code, which then also contains a defect. This illustrates that:

**A.** Failures always appear first in the documentation
**B.** Defects in early work products, if undetected, often lead to defective artifacts later in the lifecycle
**C.** Requirements cannot contain defects, only code can
**D.** The root cause of any code defect is the compiler

**5.** 'Testing shows the presence, not the absence of defects' means that:

**A.** If testing finds no defects, the software is defect-free
**B.** Testing reduces the probability that defects remain undiscovered but cannot prove correctness
**C.** Defects are only present when testing finds them
**D.** Testing must continue until all defects are found

**6.** Test conditions are defined and prioritized together with their related risks during:

**A.** Test planning
**B.** Test analysis
**C.** Test implementation
**D.** Test completion

**7.** During test implementation, testers create or acquire testware and:

**A.** Execute the test cases in the planned order
**B.** Organize test cases into test procedures and suites, prioritize them into an execution schedule, and build the test environment
**C.** Analyze the test basis to identify coverage items
**D.** Consolidate lessons learned into the completion report

**8.** Which is a DRAWBACK of high independence of testing?

**A.** It brings an outside view that authors cannot provide
**B.** Possible isolation from the development team and longer feedback loops
**C.** It challenges hidden assumptions in specifications
**D.** It always increases the number of defects found

**9.** Compared with sequential models, iterative and incremental development typically requires:

**A.** Less regression testing because changes are smaller
**B.** Fast feedback and extensive regression testing, with static and dynamic testing in each iteration
**C.** A single final system test before release
**D.** No static testing, because documentation is unnecessary

**10.** In Test-Driven Development (TDD), the working cycle is:

**A.** Design, code, test, document
**B.** Write the tests first, write the code to satisfy them, then refactor tests and code
**C.** Write acceptance criteria, automate them, then hold a specification workshop
**D.** Code first, then write tests that document the behavior

**11.** Which of the following is a RISK or challenge of adopting DevOps?

**A.** Slower feedback on code quality
**B.** The delivery pipeline must be defined and established, CI/CD tools must be introduced and maintained, and automation needs extra resources
**C.** Regression risk always increases with automation
**D.** Non-functional characteristics receive less attention

**12.** Completing static analysis of source code before dynamic testing is an example of:

**A.** Confirmation testing
**B.** The shift-left approach
**C.** Retrospective action
**D.** Test control

**13.** Which test level focuses on the behavior and capabilities of the ENTIRE system, often including end-to-end tasks?

**A.** Component testing
**B.** System testing
**C.** System integration testing
**D.** Acceptance testing

**14.** Testing that evaluates attributes other than functional characteristics - for example usability and security - is called:

**A.** Black-box testing
**B.** Non-functional testing
**C.** Confirmation testing
**D.** Maintenance testing

**15.** Testing data archiving and restore procedures when an application reaches end of life belongs to maintenance testing triggered by:

**A.** Modification
**B.** Migration
**C.** Retirement
**D.** Confirmation

**16.** Static analysis differs from dynamic testing because static analysis:

**A.** Requires executing the code at least once
**B.** Can identify problems before dynamic testing, often with less effort, and needs no test cases
**C.** Can only be applied to executable work products
**D.** Measures performance efficiency directly

**17.** During which review activity does every reviewer assess the work product and log anomalies, recommendations, and questions?

**A.** Planning
**B.** Review initiation
**C.** Individual review
**D.** Fixing and reporting

**18.** Who ensures the effective running of review meetings, including mediation, time management, and a safe environment?

**A.** Manager
**B.** Author
**C.** Moderator
**D.** Review leader

**19.** The main objective of an inspection is to:

**A.** Educate reviewers about the work product
**B.** Find the maximum number of anomalies
**C.** Generate new ideas for the author's motivation
**D.** Approve the work product for the customer

**20.** Equivalence partitions must be:

**A.** Overlapping and ordered
**B.** Non-overlapping and non-empty
**C.** Finite and continuous
**D.** Identical in size

**21.** A system accepts a country code of exactly 2 letters. Using EP, which set of test inputs covers all partitions with the fewest cases?

**A.** 'EG', 'eg', 'E1', 'Egypt'
**B.** 'EG' (valid), 'E' (too short), 'EGY' (too long), '12' (non-letters)
**C.** 'EG' and 'US' only
**D.** 'EG', '', 'EG EG'

**22.** A temperature range is defined as valid from -10.0 to +40.0 degrees. In 2-value BVA, the coverage items are:

**A.** -10.0, 0.0, +40.0
**B.** -10.1, -10.0, +40.0, +40.1
**C.** -11.0, -10.0, +39.0, +40.0
**D.** -10.0, +40.0 only

**23.** 3-value BVA is considered more rigorous than 2-value BVA because:

**A.** It tests more partitions of invalid inputs
**B.** It adds the second neighbors, catching defects that shifted-boundary implementations may hide from 2-value tests
**C.** It replaces the need for equivalence partitioning
**D.** It can be applied to unordered partitions

**24.** A rule has 4 Boolean conditions with all combinations feasible. How many rules does the full decision table have?

**A.** 4
**B.** 8
**C.** 16
**D.** 24

**25.** In a decision table, a rule's condition cell shows 'N/A'. This means:

**A.** The condition is irrelevant to the outcome
**B.** The condition is infeasible for this rule
**C.** The condition is false
**D.** The action should not occur

**26.** A test case in state transition testing is usually represented as:

**A.** A list of states to visit, one per test case
**B.** A sequence of events resulting in a sequence of state changes, usually covering several transitions
**C.** A single event with its guard condition and action
**D.** The complete state table executed column by column

**27.** Which state transition coverage criterion is the MINIMUM requirement for mission and safety-critical software?

**A.** All states coverage
**B.** Valid transitions coverage
**C.** All transitions coverage
**D.** Each choice coverage

**28.** A component has 12 executable statements and 4 branches. Test suite T1 executes 9 statements and 2 branches. What are T1's statement and branch coverage?

**A.** 75% statement, 50% branch
**B.** 9% statement, 2% branch
**C.** 75% statement, 66% branch
**D.** 50% statement, 50% branch

**29.** 100% statement coverage does NOT guarantee that:

**A.** Every executable statement was executed
**B.** All decision logic and branches were tested
**C.** A defective statement was executed
**D.** The coverage measure can be computed

**30.** A tester maintains a list of common errors - such as 'parameters passed in the wrong order' - and designs tests to expose them. This is:

**A.** Checklist-based testing
**B.** Fault attacks as a methodical form of error guessing
**C.** Session-based exploratory testing
**D.** Decision table testing

**31.** In session-based exploratory testing, the tester works within a defined time-box guided by:

**A.** The regression suite
**B.** A test charter containing test objectives, followed by a debriefing
**C.** The decision table
**D.** The exit criteria of system testing

**32.** A user story says: 'As a premium member, I want to download my invoices as PDF, so that I can archive them.' The acceptance criteria attached to the story serve to:

**A.** Replace the conversation with the business
**B.** Define the story's scope, build consensus, describe positive and negative scenarios, and serve as the basis for its acceptance testing
**C.** Document the developer's implementation plan
**D.** Prioritize the story inside the product backlog

**33.** Which set of properties makes a user story 'INVEST'?

**A.** Independent, Negotiable, Valuable, Estimable, Small, Testable
**B.** Integrated, New, Verified, Estimated, Stable, Traceable
**C.** Isolated, Named, Valued, Executable, Short, Timed
**D.** Instant, Negotiated, Voted, Evaluated, Signed, Tested

**34.** Which of the following is typical EXIT criteria content?

**A.** All smoke tests have passed
**B.** Planned tests executed, static testing performed, all found defects reported, achieved coverage level reached
**C.** Test basis and testable user stories are available
**D.** People, tools, environments, and budget are in place

**35.** Wideband Delphi differs from Planning Poker in that:

**A.** Delphi is metrics-based; Planning Poker is metrics-based too

**B.** Delphi is the general iterative expert technique; Planning Poker is its Agile variant using cards with numbers

**C.** Delphi requires historical project ratios; Planning Poker does not

**D.** Planning Poker forbids discussion between estimates

**36.** Test cases A (highest coverage), B (medium), C (low) are ordered by additional coverage prioritization. After A runs, which case runs next?

**A.** C, to complete the schedule quickly

**B.** B, the case achieving the highest ADDITIONAL coverage after A

**C.** The case with the highest risk only

**D.** Order is irrelevant once A has run

**37.** Which pair of (quadrant, example) is correct?

**A.** Q1 - user acceptance testing

**B.** Q2 - functional tests, examples, API testing, simulations

**C.** Q3 - smoke tests

**D.** Q4 - usability testing

**38.** Which response option is part of product risk control?

**A.** Ignoring risks below the quantitative threshold without documentation

**B.** Risk mitigation by testing, risk acceptance, risk transfer, or a contingency plan

**C.** Deleting the risk from the register once testing passes

**D.** Doubling the schedule for every high risk

**39.** A defect report lists: severity 'critical', priority to fix 'low', status 'deferred'. What does this combination tell the team?

**A.** The report is invalid - severity always equals priority

**B.** Impact on stakeholders is high but the fix has been consciously postponed; severity and priority are independent fields

**C.** The defect was closed without a fix

**D.** The defect was found by static testing

**40.** A team adopts containers to standardize deployment of identical test environments. Under which tool category does this fall?

**A.** Collaboration tools

**B.** Tools supporting scalability and deployment standardization

**C.** Static testing tools

**D.** Test design and implementation tools

**41.** Which pair contains one BENEFIT and one RISK of test automation?

**A.** Benefit: fewer human errors through repeatability; Risk: the open-source tool may be abandoned with no further updates
**B.** Benefit: guarantees all defects are found; Risk: slower time to market
**C.** Benefit: removes the need for test design; Risk: incompatible with manual testing
**D.** Benefit: eliminates maintenance effort; Risk: cheaper manual testing

# Part 2 - Mock Exam 2 (40 questions, 60 minutes)

Simulate the real exam completely: 40 questions, 60 minutes on the clock, closed book, no pauses, no phone. Answer every question - there is no negative marking - and flag anything that takes longer than ninety seconds for a second pass. Questions appear in syllabus order; the distribution matches the official CTFL v4.0 exam: 8 questions from Chapter 1, 7 from Chapter 2, 4 from Chapter 3, 13 from Chapter 4, 6 from Chapter 5, and 2 from Chapter 6. Record your answers on paper, then score with the key in Part 4. Pass mark: 26 of 40 (65 percent).

**1.** Which of the following is a typical test objective?

**A.** Verifying compliance with contractual, legal, and regulatory requirements
**B.** Proving the absence of defects to certify the product
**C.** Executing every possible input combination at least once
**D.** Assigning blame for each failure to its author

**2.** Checking whether the system meets specified requirements is called:

**A.** Validation
**B.** Verification
**C.** Regression testing
**D.** Root cause analysis

**3.** A static testing activity identifies 5 defects in a design document. After fixes, what further activity is needed and why?

**A.** Failure reproduction, because every defect needs a documented failure
**B.** Only removal of the defects; static findings require no failure reproduction or diagnosis
**C.** Regression testing of the design document against the code
**D.** Nothing; static defects disappear once reported

**4.** Which statement about errors, defects, and failures is correct?

**A.** Every defect always causes a failure when executed
**B.** Some defects result in failures only in specific circumstances, and some may never cause a failure
**C.** Failures can only originate from defects in source code
**D.** Errors are always introduced by tools, not people

**5.** A root cause analysis after a production failure shows operators were not trained on a new screen. Removing the training gap is an example of:

**A.** Risk acceptance

**B.** Addressing the root cause to prevent or reduce similar future failures

**C.** Confirmation testing

**D.** Test control via new directives

**6.** Test design, test data requirements, test environment design, and identification of coverage items happen during:

**A.** Test analysis

**B.** Test design

**C.** Test execution

**D.** Test monitoring and control

**7.** Test completion activities occur:

**A.** Only when the whole project is cancelled

**B.** At project milestones such as a completed test level, a finished agile iteration, a release, or a completed maintenance release

**C.** Only after the customer signs acceptance

**D.** Only when exit criteria cannot be met

**8.** In the whole-team approach, which statement about quality responsibility is correct?

**A.** Quality remains the testers' exclusive responsibility

**B.** The whole team - testers, developers, and business representatives - is responsible for quality

**C.** Quality transfers entirely to the product owner

**D.** Responsibility is defined by the independent test team's charter

**9.** In sequential development models, testers in the initial phases typically:

**A.** Execute dynamic tests on partial builds

**B.** Participate in requirement reviews, test analysis, and test design, because executable code comes later

**C.** Prepare only the production deployment

**D.** Skip all documentation because specifications change

**10.** Agile projects favor lightweight documentation and extensive test automation because:

**A.** Documentation is always harmful to quality

**B.** Change is assumed throughout the project and automation keeps regression testing affordable

**C.** Manual testing is prohibited in Agile teams

**D.** Agile teams must avoid all experience-based techniques

**11.** ATDD derives tests from:

**A.** Unit test frameworks

**B.** Acceptance criteria as part of the system design process, before the application part is developed

**C.** The deployed production logs

**D.** The original waterfall plan

**12.** Which statement about retrospectives is correct?

**A.** Only testers may attend retrospectives

**B.** Results should be recorded and are normally part of the test completion report; recommended improvements must be followed up

**C.** Retrospectives are held only at the end of the final release

**D.** Retrospectives replace configuration management

**13.** Acceptance testing ideally performed by the intended users to validate business needs is called:

**A.** Operational acceptance testing

**B.** User acceptance testing (UAT)

**C.** Contractual acceptance testing

**D.** Alpha testing

**14.** White-box testing derives tests from:

**A.** The specification external to the test object

**B.** The system's implementation or internal structure, such as code, architecture, workflows, and data flows

**C.** User feedback from beta testing

**D.** The risk register only

**15.** A hot fix was deployed last night. Which testing MUST accompany such unplanned maintenance?

**A.** Only re-running the full acceptance test suite

**B.** Evaluating the fix itself and regression checking the unchanged parts of the system, sized by impact analysis

**C.** No testing, because hot fixes are pre-approved

**D.** Only static analysis of the patch

**16.** For which work product is static ANALYSIS (tool-based) appropriate but a review alone insufficient?

**A.** A contract reviewed by lawyers

**B.** Source code with a formal syntax checked by a tool

**C.** A user story discussed in refinement

**D.** A test charter reviewed by the team

**17.** What does review INITIATION aim to achieve?

**A.** Defining the review scope and exit criteria
**B.** Making sure everyone and everything is prepared: access to the work product, clear roles, and everything needed to perform the review
**C.** Deciding the quality level of the work product
**D.** Creating defect reports for corrective action

**18.** Who takes OVERALL responsibility for a review, such as deciding who is involved and organizing when and where it takes place?

**A.** Moderator
**B.** Review leader
**C.** Manager
**D.** Scribe

**19.** A team wants a review that builds consensus on a technical decision with technically qualified reviewers led by a moderator. The appropriate type is:

**A.** Informal review
**B.** Walkthrough
**C.** Technical review
**D.** Inspection

**20.** Partitions can be identified for which data elements related to the test object?

**A.** Inputs only
**B.** Inputs, outputs, configuration items, internal values, time-related values, and interface parameters
**C.** Only numeric input fields
**D.** Only fields with a defined range

**21.** A flight booking accepts passengers aged 2 to 99. With multiple test objects each having their own partition sets, 'Each Choice coverage' requires:

**A.** Every combination of partitions to be covered
**B.** Test cases to exercise each partition from each set at least once, without regard to combinations
**C.** Only boundary values of each set
**D.** One test case per test object

**22.** For the range 1-500, which values does 3-VALUE BVA add compared with 2-value BVA (0, 1, 500, 501)?

**A.** 2 and 499
**B.** -1 and 502
**C.** 250 and 251
**D.** Nothing; both variants test identical values

**23.** In 3-value BVA, some coverage items:

**A.** Are always invalid partition values
**B.** May not be boundary values, because they include both neighbors of each boundary
**C.** Must be excluded from the coverage percentage
**D.** Are drawn from the state table

**24.** A decision table is 'minimized' by:

**A.** Deleting all invalid partitions
**B.** Merging columns in which some conditions do not affect the outcome into a single column
**C.** Doubling the number of rules
**D.** Replacing Boolean conditions with extended entries

**25.** The strength of decision table testing is that it:

**A.** Guarantees branch coverage of the code
**B.** Provides a systematic approach to identify all combinations of conditions and helps find gaps or contradictions in requirements
**C.** Replaces the need for risk-based testing
**D.** Is the fastest technique for very large numbers of conditions

**26.** In a state table, invalid transitions are represented by:

**A.** Arrows labeled with guard conditions
**B.** Empty cells, which the diagram does not show explicitly
**C.** Rows without states
**D.** Double-headed arrows

**27.** A test suite visits every state and attempts every valid AND invalid transition. Which claim is justified?

**A.** Only all states coverage is achieved
**B.** All transitions coverage is achieved, which also guarantees full all states and full valid transitions coverage
**C.** Valid transitions coverage only, because invalid attempts do not count
**D.** No criterion is achieved until the state table is complete

**28.** A code fragment has 10 branches. Test suite T2 exercises 8 of them. Branch coverage is:

**A.** 80%
**B.** 8%
**C.** 12.5%
**D.** 100% of statements executed

**29.** Why does white-box testing remain useful when the specification is vague, outdated, or incomplete?

**A.** Because it ignores the implementation entirely

**B.** Because the entire software implementation is taken into account, facilitating defect detection even without a good specification

**C.** Because it measures requirement coverage directly

**D.** Because it tests only the documented behavior

**30.** Errors, defects, and failures are generally related to which areas according to the error guessing guidance?

**A.** Input, output, logic, computation, interfaces, and data

**B.** Budget, schedule, scope, and suppliers

**C.** Reviews, walkthroughs, technical reviews, inspections

**D.** Planning, monitoring, control, completion

**31.** Which statement about checklists is correct?

**A.** Checklists should contain every possible check, including automated ones

**B.** Checklist items are often phrased as questions, checkable separately and directly, and the checklist should be regularly updated from defect analysis without becoming too long

**C.** Once written, a checklist remains effective forever

**D.** High-level checklists guarantee repeatability of testing

**32.** In the Given/When/Then format used for scenario-oriented acceptance criteria, the sections correspond to:

**A.** Role, goal, business value

**B.** Precondition/state, event/action, expected outcome

**C.** Card, conversation, confirmation

**D.** Entry criteria, action, exit criteria

**33.** Which entry criterion example is correct for starting test execution?

**A.** All regression tests are automated

**B.** Test environment is built and verified, test data is available, and smoke tests pass on the build

**C.** Defect density is below the agreed threshold

**D.** The completion report has been distributed

**34.** Extrapolation as an estimation technique is especially suitable for iterative SDLCs because:

**A.** It ignores all historical data

**B.** Measurements made early in the current project can be used to project the remaining effort, e.g. averaging the last three iterations

**C.** It requires a signed contract

**D.** It is the only technique approved in Agile

**35.** Using three-point estimation with a = 12, m = 16, b = 30 person-hours, E and SD equal:

**A.** E = 17.7, SD = 3
**B.** E = 16, SD = 18
**C.** E = 19.3, SD = 3
**D.** E = 18, SD = 6

**36.** The team's velocity drops after two testers fall ill. Under which risk category does this fall?

**A.** Product risk
**B.** Project risk - people issues
**C.** Product risk - usability
**D.** Environmental maintenance trigger

**37.** In the qualitative approach to risk assessment, the risk level is determined using:

**A.** The multiplication of likelihood and impact
**B.** A risk matrix
**C.** The test pyramid
**D.** Defect density metrics

**38.** Which communication option suits DISTRIBUTED teams that cannot rely on face-to-face contact?

**A.** Only verbal communication in hallways
**B.** More formal communication channels such as dashboards, electronic channels, and formal reports
**C.** No communication, to avoid interruptions
**D.** Exclusive reliance on memory and tribal knowledge

**39.** A test manager needs statistics, graphs, and aggregated data about test progress and defect rates for reporting. Which tool support fits BEST?

**A.** Static testing tools
**B.** Test execution tools feeding reporting/management tooling - automation eases access to such information
**C.** Containerization tools
**D.** Spelling checkers

**40.** Choosing a test tool that is incompatible with the development platform is an example of:

**A.** A benefit of automation
**B.** A risk of test automation
**C.** A test level decision
**D.** An exit criterion violation

# Part 4 - Mock Exam Answer Keys

Score each mock by comparing your recorded answers with the key below, one point per correct answer. Then read the rationale for every question you missed or guessed - the explanations quote the syllabus rule that decides the question. Classify each miss as a vocabulary gap, a distinction mix-up, or a calculation slip, and repair accordingly using the matching chapter of the Study Notes.

## Mock Exam 1 - Answer Grid

| # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q 1 | B | Q 2 | B | Q 3 | B | Q 4 | B | Q 5 | B | Q 6 | B | Q 7 | B | Q 8 | B | Q 9 | B | Q 10 | B |
| Q 11 | B | Q 12 | B | Q 13 | B | Q 14 | B | Q 15 | C | Q 16 | B | Q 17 | C | Q 18 | C | Q 19 | B | Q 20 | B |
| Q 21 | B | Q 22 | B | Q 23 | B | Q 24 | C | Q 25 | B | Q 26 | B | Q 27 | C | Q 28 | A | Q 29 | B | Q 30 | B |
| Q 31 | B | Q 32 | B | Q 33 | A | Q 34 | B | Q 35 | B | Q 36 | B | Q 37 | B | Q 38 | B | Q 39 | B | Q 40 | B |

## Mock Exam 2 - Answer Grid

| # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans | # | Ans |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q 1 | A | Q 2 | B | Q 3 | B | Q 4 | B | Q 5 | B | Q 6 | B | Q 7 | B | Q 8 | B | Q 9 | B | Q 10 | B |
| Q 11 | B | Q 12 | B | Q 13 | B | Q 14 | B | Q 15 | B | Q 16 | B | Q 17 | B | Q 18 | B | Q 19 | C | Q 20 | B |
| Q 21 | B | Q 22 | A | Q 23 | B | Q 24 | B | Q 25 | B | Q 26 | B | Q 27 | B | Q 28 | A | Q 29 | B | Q 30 | A |
| Q 31 | B | Q 32 | B | Q 33 | B | Q 34 | B | Q 35 | A | Q 36 | B | Q 37 | B | Q 38 | B | Q 39 | B | Q 40 | B |

## Mock Exam 1 - Rationales

| Q | Ans | Rationale |
|---|---|---|
| 1 | B | Testing includes more than execution: planning, analysis, design, implementation, static testing, and reporting. It involves verification AND validation. |

| Q | Ans | Rationale |
|---|---|---|
| 2 | B | Validation checks the system meets users' and stakeholders' needs in the operational environment; verification checks it meets the specified requirements. |
| 3 | B | Testing triggers failures (dynamic) or finds defects directly (static); debugging reproduces, diagnoses, and fixes. Static findings need no reproduction. |
| 4 | B | Undetected early defects cascade into later artifacts - a key argument for early testing (principle 3). |
| 5 | B | Principle 1: finding no defects proves nothing; testing cannot establish absence of defects. |
| 6 | B | Test analysis identifies testable features and defines and prioritizes test conditions with related risks; it answers what to test. |
| 7 | B | Test implementation creates testware, assembles procedures/suites, schedules execution, and sets up and verifies the environment. Execution follows. |
| 8 | B | Independence benefits (outside view, objective reporting, challenging assumptions) come with drawbacks: isolation, slower feedback, and possible loss of developer ownership. |
| 9 | B | Each iteration delivers an increment, so both static and dynamic testing occur at all levels each cycle, and frequent delivery demands fast feedback plus heavy regression. |
| 10 | B | TDD directs coding through test cases: tests first, code to satisfy them, then refactoring of both. |
| 11 | B | DevOps challenges: pipeline definition, tooling introduction and maintenance, and the resource cost of establishing automation. |
| 12 | B | Shift-left practices include early static analysis, specification reviews, tests before code, CI/CD, and early non-functional testing. |
| 13 | B | System testing focuses on overall behavior and capabilities of the whole system, both functional end-to-end and non-functional, often by an independent team. |
| 14 | B | Non-functional testing evaluates how well the system behaves, per ISO/IEC 25010 characteristics such as usability, security, performance efficiency. |
| 15 | C | Retirement triggers include testing data archiving when long retention is required, and restore/retrieval procedures during the retention period. |
| 16 | B | Static analysis works without execution and without test cases, often inside CI, detecting code defects and evaluating maintainability and security. |
| 17 | C | Individual review is where reviewers apply techniques such as checklist-based or scenario-based reviewing and log their findings. |
| 18 | C | The moderator (facilitator) runs the meetings effectively and keeps the environment safe for everyone to speak freely. |
| 19 | B | Inspection's main objective is finding the maximum number of anomalies; other objectives include quality evaluation, confidence, and author improvement, with metrics collected. |
| 20 | B | Partitions must not overlap and must be non-empty sets; they may be continuous or discrete, ordered or unordered, finite or infinite. |
| 21 | B | One value per partition: the valid 2-letter partition plus each invalid partition (too short, too long, non-letter characters) - three invalid + one valid = 4 cases. |

| Q | Ans | Rationale |
|---|---|---|
| 22 | B | 2-value BVA: each boundary plus its closest neighbor in the adjacent partition: -10.1, -10.0, +40.0, +40.1. Adding second neighbors (-11.0, +39.0...) would be 3-value BVA. |
| 23 | B | 3-value BVA includes both neighbors of each boundary; some coverage items are not boundary values, catching defects such as x <= 10 implemented as x == 10 (value 9 exposes it). |
| 24 | C | Full table columns = 2^n = 2^4 = 16. Minimization or risk-based selection is used when this becomes too expensive. |
| 25 | B | N/A marks an infeasible condition for the rule; a dash marks an irrelevant condition; T/F give satisfied/not satisfied. |
| 26 | B | A state transition test case is a sequence of events producing a sequence of state changes (and actions); one case typically covers several transitions. |
| 27 | C | Full all-transitions coverage (valid plus attempted invalid transitions) should be a minimum requirement for mission and safety-critical software. |
| 28 | A | Statement coverage = 9/12 = 75%; branch coverage = 2/4 = 50%. Coverage is always exercised items over total items. |
| 29 | B | Statement coverage ensures all statements ran, including defective ones, but may not exercise all branches; branch testing covers decision outcomes. |
| 30 | B | Fault attacks are the methodical implementation of error guessing: create/acquire lists of possible errors and design tests that expose the defects. |
| 31 | B | Session-based exploratory testing uses a time-box, a charter with objectives, session sheets, and a debriefing with stakeholders. |
| 32 | B | Acceptance criteria define scope, reach consensus, describe positive and negative scenarios, enable planning and estimation, and form the basis for acceptance testing. |
| 33 | A | INVEST = Independent, Negotiable, Valuable, Estimable, Small, Testable (Wake 2003). |
| 34 | B | Exit criteria are thoroughness measures (coverage, unresolved defects, defect density, failed cases) and completion criteria (planned tests run, static testing done, defects reported). The others are entry criteria. |
| 35 | B | Wideband Delphi: isolated expert estimates, discussion of deviations, re-estimation until consensus. Planning Poker is the Agile card-based variant. |
| 36 | B | In additional coverage prioritization, each subsequent test case is the one achieving the highest additional coverage over those already run. |
| 37 | B | Q2 (business facing, support the team) holds functional tests, examples, story tests, UX prototypes, API testing, simulations. Q1: component/integration in CI; Q3: exploratory, usability, UAT; Q4: smoke and non-functional except usability. |
| 38 | B | Response options after analysis: mitigation by testing, acceptance, transfer, contingency. Mitigation by testing includes right-skilled testers, independence, reviews/static analysis, techniques and coverage, relevant test types, dynamic and regression testing. |
| 39 | B | Severity (degree of impact) and priority (urgency to fix) are separate fields; statuses such as deferred or duplicate are part of the defect workflow. |
| 40 | B | Virtual machines and containerization tools support scalability and deployment standardization of environments. |

| Q | Ans | Rationale |
|---|---|---|
| 41 | B | Benefits include consistency/repeatability and prevention of simple human errors; risks include open-source abandonment, vendor dependency, and unrealistic expectations. |

## Mock Exam 2 - Rationales

| Q | Ans | Rationale |
|---|---|---|
| 1 | A | Objectives include verifying compliance with contractual/legal/regulatory requirements, evaluating work products, triggering failures, reducing risk, informing stakeholders, building confidence, and validating completeness. |
| 2 | B | Verification checks the system meets specified requirements; validation checks it meets users' needs in the operational environment. |
| 3 | B | With static testing there is no failure to reproduce or diagnose; debugging is limited to removing the defect directly. |
| 4 | B | Some defects always fail when executed, some fail only under specific conditions, and some never fail. Failures can also stem from environmental causes. |
| 5 | B | Root causes are identified through root cause analysis when failures occur; addressing them (here, training) prevents or reduces similar failures. |
| 6 | B | Test design elaborates test conditions into test cases and other testware, identifies coverage items, defines test data requirements, and designs the environment - it answers how to test. |
| 7 | B | Test completion collects data at milestones: level completed, iteration finished, project completed or cancelled, system released, maintenance release completed. |
| 8 | B | The whole-team approach makes quality everyone's responsibility, leveraging all skill sets and improving communication. |
| 9 | B | Sequential models create code late, so early tester work concentrates on reviews, analysis, and design; dynamic testing comes later. |
| 10 | B | Agile assumes constant change; lightweight docs plus automation make frequent regression feasible, and manual testing leans on experience-based techniques. |
| 11 | B | ATDD derives tests from acceptance criteria during system design, written before the code that satisfies them. |
| 12 | B | Retrospectives discuss successes, failures, and improvements; results are recorded and normally join the test completion report, and improvements must be followed up. |
| 13 | B | UAT is validation by intended users of business needs; operational acceptance covers operations processes; contractual/regulatory follow contracts or standards; alpha and beta are access-based forms. |
| 14 | B | White-box testing is structure-based, derived from implementation/structure; black-box is specification-based. |
| 15 | B | Maintenance testing evaluates the success of the change and checks for regressions in unchanged parts (usually most of the system); impact analysis helps size it. |
| 16 | B | Static analysis needs a structure against which to check: code with formal syntax, models, or formatted text. Reviews work on anything readable. |

| Q | Ans | Rationale |
|---|---|---|
| 17 | B | Review initiation ensures every participant has access, understands their role and responsibilities, and receives everything needed. Scope definition belongs to planning. |
| 18 | B | The review leader takes overall responsibility for the review; the manager decides what is reviewed and funds it; the moderator runs the meeting. |
| 19 | C | Technical reviews are performed by technically qualified reviewers, led by a moderator, aiming at consensus and decisions on technical problems. |
| 20 | B | EP partitions can be built for any data element: inputs, outputs, configuration items, internal values, time-related values, interface parameters. |
| 21 | B | Each Choice coverage requires each partition of each set to be exercised at least once; it does not take combinations into account. |
| 22 | A | 3-value BVA adds the second neighbors: 2 (above 1) and 499 (below 500), giving 0, 1, 2, 499, 500, 501. |
| 23 | B | 3-value BVA takes each boundary plus both neighbors; the second neighbors are not boundary values themselves. |
| 24 | B | Minimization merges columns whose conditions do not affect the outcome; simplification deletes infeasible columns. Minimization algorithms are out of syllabus scope. |
| 25 | B | Decision tables systematically expose condition combinations, including overlooked ones, and reveal gaps or contradictions; with many conditions rules grow exponentially (2^n). |
| 26 | B | The state table shows invalid transitions as empty cells - an advantage over the state transition diagram, which shows only valid transitions. |
| 27 | B | Full all-transitions coverage guarantees full valid-transitions and full all-states coverage - the strictest criterion of the three. |
| 28 | A | Branch coverage = exercised branches / total branches = 8/10 = 80%. |
| 29 | B | White-box strength: the whole implementation is examined, so defects surface even with poor specifications; the weakness is missing defects of omission. |
| 30 | A | Error guessing domains: input, output, logic, computation, interfaces, data - e.g. correct input not accepted, wrong format, missing cases, parameter mismatch, incorrect initialization. |
| 31 | B | Checklist items are question-form, individually checkable; exclude automated checks, entry/exit criterion items, and over-general items; update regularly but keep short. High-level checklists reduce repeatability. |
| 32 | B | Given sets the state/precondition, When gives the event or action, Then describes the expected outcome - the BDD format also cited for acceptance criteria. |
| 33 | B | Entry criteria: resources available (environment, data, people), testware available (basis, cases), initial quality level such as passing smoke tests. The others resemble exit criteria or completion. |
| 34 | B | Extrapolation uses measurements from the current project as early as possible and a mathematical model to approximate remaining effort - a natural fit for iterations. |
| 35 | A | E = (12 + 64 + 30) / 6 = 106 / 6 = 17.67 ≈ 17.7; SD = (30 - 12) / 6 = 3. |
| 36 | B | People issues such as insufficient staff, conflicts, and skills gaps are project risks; product risks concern the product's quality characteristics. |

| Q | Ans | Rationale |
|---|---|---|
| 37 | B | Qualitative assessment reads the risk level from a risk matrix; the quantitative approach multiplies likelihood by impact. |
| 38 | B | More formal communication is often appropriate for distributed teams; dashboards, electronic channels, documentation, and formal reports keep status visible. |
| 39 | B | Automation provides easier access to information about testing - statistics, graphs, aggregated progress, defect rates, execution durations - supporting management and reporting. |
| 40 | B | Platform incompatibility is one of the listed risks of test automation, alongside vendor dependency, unrealistic expectations, and unsuitable compliance choices. |

### After scoring

Finished both mocks at 32 or above? Your vocabulary and your calculation speed are in passing territory. Spend the final day on the Cheat Sheet and your error notebook, and walk in calm. Below target? Re-drill the two chapters with the most misses - the distribution means Chapter 4 repairs pay the most.